

AN AI POWERED SOFTWARE-BASED FIRE DETECTION SYSTEM FOR CCTV PLATFORMS.

FiremeX - TEST CASES

OPERATOR MANUAL TESTING

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Project	FiremeX
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1. Purpose

This document contains detailed UI test cases for validating the core dashboard, fire/smoke detection, alerting, and livefeed pages of the FiremeX system. Each test case is traceable to a system requirement.

2. Test Case Details

TC ID	Title	Precondition	Test Steps	Expected Result	Priority	Req ID
Dashboard - Module Test Cases						
TC_DB_001	Notification icon is clickable	User is logged in as Operator and is on the Dashboard page.	1. Locate the notification (bell) icon in the top bar. 2. Click the notification icon.	Notification icon responds to the click and the notifications panel/dropdown opens.	High	DB-01
TC_DB_002	Notification icon shows unviewed count	User is logged in as Operator and is on the Dashboard page.	1. Trigger one or more new notifications. 2. Observe the notification icon.	A badge on the notification icon displays the correct count of unviewed notifications and updates as new notifications arrive.	High	DB-01
TC_DB_003	Side bar options are clickable	User is logged in as Operator and is on the Dashboard page.	1. Click on each side bar menu option (Dashboard, Live feed, Incidents, Alerts, Settings) in turn.	Each side bar option is clickable and navigates the operator to the corresponding page.	High	DB-02
TC_DB_004	Active side bar tab is highlighted	User is logged in as Operator and is on the Dashboard page.	1. Click on any side bar tab. 2. Observe the visual state of the clicked tab.	The clicked/active tab is visually highlighted to indicate the currently active page.	Medium	DB-02
TC_DB_005	User role and user name are visible	User is logged in as Operator and is on the Dashboard page.	1. Look at the top bar / side bar user profile section.	The logged-in user's role and name are correctly displayed.	Medium	DB-02
TC_DB_006	User profile icon is clickable	User is logged in as Operator and is on the Dashboard page.	1. Click on the user profile icon in the top/side bar.	The user profile icon responds to the click and shows the profile menu/options.	Medium	DB-02
TC_DB_007	Static cards details are visible	User is logged in as Operator and is on the Dashboard page.	1. Navigate to the Dashboard page. 2. Observe the static summary cards.	All static card details (labels and values) are visible and correctly populated.	High	DB-03
TC_DB_008	Static cards use correct colour theory	User is logged in as Operator and is on the Dashboard page.	1. Observe the colour coding applied on each static card.	Each static card displays the correct colour based on the defined colour theory (e.g. status/severity based colours).	Low	DB-03
TC_DB_009	'View all' link in Recent Incident section is clickable	User is logged in as Operator and is on the Dashboard page.	1. Locate the Recent Incident section. 2. Click on the 'View all' text/link.	Operator is navigated to the full Incidents page/list.	Medium	DB-04

TC_DB_010	Recent Incident section shows only 5 most recent alerts	User is logged in as Operator and is on the Dashboard page.	1. Ensure more than 5 incidents/alerts exist in the system. 2. Observe the Recent Incident section on the Dashboard.	Only the 5 most recent alerts are listed in the Recent Incident section.	High	DB-04
TC_DB_011	Recent Incident section shows required columns	User is logged in as Operator and is on the Dashboard page.	1. Observe the Recent Incident section table/list.	The section displays Time, Camera, Zone, AI Confidence, and Status for each of the recent alerts.	High	DB-04
TC_DB_012	Recent Camera Preview shows correct detection status	User is logged in as Operator and is on the Dashboard page.	1. Trigger a Fire detection and a Smoke detection event. 2. Observe the Recent Camera Preview section.	The preview correctly shows the status as 'Fire detected' or 'Smoke detected' as applicable for each camera.	High	DB-05
TC_DB_013	Recent Camera Preview shows camera number and zone	User is logged in as Operator and is on the Dashboard page.	1. Observe a camera preview tile in the Recent Camera Preview section.	The preview displays the correct camera number and camera zone.	Medium	DB-05
TC_DB_014	All detected cameras are shown in Recent Camera Preview	User is logged in as Operator and is on the Dashboard page.	1. Trigger detections on multiple cameras. 2. Observe the Recent Camera Preview section.	All cameras with an active detection are shown in this section.	High	DB-05
TC_DB_015	Side bar Dashboard tab is highlighted	User is logged in as Operator and is on the Dashboard page.	1. Navigate to the Dashboard page via the side bar.	The Dashboard tab in the side bar is highlighted, confirming the current page.	Medium	DB-06

Live feed - Module Test Cases

TC_LF_001	2x2 layout button is clickable	User is logged in as Operator and is on the Live feed page.	1. Click on the '2x2' layout button.	The live feed grid switches to a 2x2 camera layout.	Medium	LF-01
TC_LF_002	1x2 layout button is clickable	User is logged in as Operator and is on the Live feed page.	1. Click on the '1x2' layout button.	The live feed grid switches to a 1x2 camera layout.	Medium	LF-01
TC_LF_003	Add Camera button is clickable	User is logged in as Operator and is on the Live feed page.	1. Click on the 'Add Camera' button.	The Camera Registration form/modal opens.	High	LF-02
TC_LF_004	Submit button is disabled until required fields are filled	User is logged in as Operator and is on the Live feed page.	1. Open the Add Camera form. 2. Leave required fields empty. 3. Observe the Submit button. 4. Fill in all required fields.	Submit button remains disabled while required fields are empty, and becomes enabled only once all required fields are filled.	High	LF-02
TC_LF_005	Cancel button on Add Camera form is clickable	User is logged in as Operator and is on the Live feed page.	1. Open the Add Camera form. 2. Click the Cancel button.	The form closes without saving any data and no new camera is registered.	Medium	LF-02
TC_LF_006	AI tracking toggle is enabled on Camera Registration	User is logged in as Operator and is on the Live feed page.	1. Open the Camera Registration form. 2. Toggle the AI tracking option.	The AI tracking button/toggle can be enabled successfully on	Medium	LF-03

				the camera registration form.		
TC_LF_007	Required fields validation on Camera Registration	User is logged in as Operator and is on the Live feed page.	1. Open the Camera Registration form. 2. Attempt to submit with missing required fields.	Form validation prevents submission and highlights the missing required fields.	High	LF-03
TC_LF_008	Input data format validation on Camera Registration	User is logged in as Operator and is on the Live feed page.	1. Open the Camera Registration form. 2. Enter invalid data formats (e.g. invalid IP address) in the input fields.	Form displays an appropriate validation error for incorrectly formatted input data.	High	LF-03
TC_LF_009	Successful camera registration confirmation	User is logged in as Operator and is on the Live feed page.	1. Fill the Camera Registration form with valid, correctly formatted data. 2. Click Submit.	A camera registration success message is displayed and the new camera's live preview appears in the Live feed section.	High	LF-03
TC_LF_010	Critical/altered camera shown in red	User is logged in as Operator and is on the Live feed page.	1. Simulate a camera going into a critical/altered (e.g. tampered or offline) state. 2. Observe the camera card.	The affected camera card is highlighted in red to indicate the critical/altered state.	High	LF-04
TC_LF_011	Edit button on camera card is clickable	User is logged in as Operator and is on the Live feed page.	1. Locate a camera card. 2. Click the Edit button.	The camera edit form opens pre-populated with that camera's existing details.	Medium	LF-04
TC_LF_012	Delete button on camera card is clickable	User is logged in as Operator and is on the Live feed page.	1. Locate a camera card. 2. Click the Delete button. 3. Confirm deletion if prompted.	The camera is removed from the Live feed page and is no longer listed.	High	LF-04
TC_LF_013	Camera card shows required details	User is logged in as Operator and is on the Live feed page.	1. Observe a camera card on the Live feed page.	The card displays Camera Number, Zone, IP Address, and Status (Online/Offline).	High	LF-04
TC_LF_014	All registered cameras are shown	User is logged in as Operator and is on the Live feed page.	1. Register multiple cameras. 2. Navigate to the Live feed page.	All registered cameras are displayed as camera cards on the Live feed page.	High	LF-04
TC_LF_015	Notification icon is clickable (Live feed page)	User is logged in as Operator and is on the Live feed page.	1. Locate the notification icon on the Live feed page. 2. Click on it.	The notifications panel/dropdown opens.	Medium	LF-05
TC_LF_016	Notification icon shows unviewed count (Live feed page)	User is logged in as Operator and is on the Live feed page.	1. Trigger a new notification. 2. Observe the notification icon on the Live feed page.	The notification icon displays the correct count of unviewed notifications.	Medium	LF-05
TC_LF_017	Each camera shows correct time stamp	User is logged in as Operator and is on the Live feed page.	1. Observe the live feed of any camera card.	Each camera tile displays the correct, current time stamp.	Medium	LF-06

TC_LF_018	Side bar Live feed tab is highlighted	User is logged in as Operator and is on the Live feed page.	1. Navigate to the Live feed page via the side bar.	The Live feed tab in the side bar is highlighted, confirming the current page.	Low	LF-07
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Incidents - Module Test Cases

TC_IN_C_001	Side bar Incidents tab is highlighted	User is logged in as Operator and is on the Incidents page.	1. Navigate to the Incidents page via the side bar.	The Incidents tab in the side bar is highlighted, confirming the current page.	Medium	INC-01
TC_IN_C_002	Camera Name filter is clickable and filters correctly	User is logged in as Operator and is on the Incidents page.	1. Click on the Camera Name filter. 2. Select a specific camera name.	The incident list is filtered to show only incidents for the selected camera.	High	INC-02
TC_IN_C_003	Date filter is clickable and filters correctly	User is logged in as Operator and is on the Incidents page.	1. Click on the Date filter. 2. Select a date/date range.	The incident list is filtered to show only incidents within the selected date/date range.	High	INC-02
TC_IN_C_004	Incident Status filter is clickable and filters correctly	User is logged in as Operator and is on the Incidents page.	1. Click on the Incident Status filter. 2. Select a status (e.g. Resolved/Unresolved).	The incident list is filtered to show only incidents matching the selected status.	High	INC-02
TC_IN_C_005	Every new incident defaults to 'Unresolved'	User is logged in as Operator and is on the Incidents page.	1. Trigger a new incident/alert. 2. Observe its status on the Incidents page.	The new incident's status is set to 'Unresolved' by default.	High	INC-03
TC_IN_C_006	Action column dropdown is clickable with correct colour coding	User is logged in as Operator and is on the Incidents page.	1. Locate the Action column for an incident. 2. Click the dropdown.	The dropdown opens, is clickable, and options are colour coded per the defined colour theory.	Medium	INC-04
TC_IN_C_007	Incident status is updatable via Action dropdown	User is logged in as Operator and is on the Incidents page.	1. Open the Action dropdown for an incident. 2. Select a new status.	The incident's status updates to the newly selected value.	High	INC-04
TC_IN_C_008	Resolving an incident requires a resolve note	User is logged in as Operator and is on the Incidents page.	1. Open the Action dropdown for an incident. 2. Select 'Resolved'.	Operator is prompted to enter a resolve note before the status change is accepted.	High	INC-04
TC_IN_C_009	Incident cannot be marked Resolved without a resolve note	User is logged in as Operator and is on the Incidents page.	1. Select 'Resolved' from the Action dropdown. 2. Attempt to save without entering a resolve note.	The system blocks the action and the incident status is not changed to 'Resolved'.	High	INC-04
TC_IN_C_010	Incident record shows all required data columns	User is logged in as Operator and is on the Incidents page.	1. Observe an incident record row on the Incidents page.	Timestamp, Camera, Zone, AI Confidence, Status, and Actions are all populated for each record.	High	INC-05

Alerts - Module Test Cases

TC_AL T_001	Static cards update dynamically	User is logged in as Operator and is on the Alerts page.	1. Trigger a new alert. 2. Observe the static summary cards on the Alerts page.	The static cards update dynamically to reflect the new alert data without requiring a manual page refresh.	High	ALT-01
TC_AL T_002	Acknowledge column button is clickable	User is logged in as Operator and is on the Alerts page.	1. Locate the Acknowledge button for an alert row. 2. Click the button.	The alert is marked as acknowledged and the button/state updates accordingly.	High	ALT-02
TC_AL T_003	Data filter is clickable and filters correctly	User is logged in as Operator and is on the Alerts page.	1. Click on the data/date filter. 2. Select a date or range.	The alerts list is filtered to show only alerts within the selected date/range.	High	ALT-02
TC_AL T_004	Channel filter is clickable and filters correctly	User is logged in as Operator and is on the Alerts page.	1. Click on the Channel filter. 2. Select a specific channel/camera.	The alerts list is filtered to show only alerts from the selected channel.	High	ALT-02
TC_AL T_005	All column details are filled for each alert	User is logged in as Operator and is on the Alerts page.	1. Observe an alert row on the Alerts page.	All expected column values are populated (no blank/missing data) for each alert record.	Medium	ALT-03
TC_AL T_006	Side bar Alerts tab is highlighted	User is logged in as Operator and is on the Alerts page.	1. Navigate to the Alerts page via the side bar.	The Alerts tab in the side bar is highlighted, confirming the current page.	Low	ALT-04

3. Notes

- Each test case's 'Actual Result' and 'Pass/Fail' status is recorded during execution.
- Priority levels: Critical = safety-impacting, High = major functional, Medium = usability/operational.